

Pratham Agrawal

+91-8269067007 — prathamagrawal1205@gmail.com — linkedin.com/in/pratham-manish-agrawal
github.com/prathamagrawal — leetcode.com/u/prathamagrawal — kaggle.com/fearsomejockey

Experience

Affinsys AI — AI B2B Startup

Software Engineer

Bangalore, India

Jan 2024 - Present

- **DBOps Service:** Automated enterprise data pipeline authoring by building an **LLM-driven ETL orchestrator** using the **Anthropic Claude API** with structured prompt chaining — enabling self-serve workflow creation for data ops teams and reducing manual pipeline authoring from hours to under 2 minutes.
- **Datalens Service:** Architected a self-serve **analytics platform enabling NLQ-to-SQL generation via LLMs**, using a **RAG pipeline over schema and query metadata** for context-aware query construction, with SQL sanitisation and a chart catalog engine supporting 15+ chart types. **Reduced dashboard creation time by 90%**, enabling non-technical stakeholders to independently build and iterate on reports.
- **Analytics Service:** Led end-to-end development of a multi-tenant analytics **service powering 20+ dashboards and 200+ charts for internal and client-facing teams**. Introduced materialized SQL views to eliminate redundant query execution, reducing database load by 40% and **cutting average API response time from 800ms to 480ms**.
- **Analytics Scheduler:** Designed a scheduling engine to **automate recurring delivery of 100+ analytics reports via email, reducing manual reporting effort by an estimated 15 engineer-hours per week**.
- **Archival Service:** Implemented a **comprehensive archival and provisioning pipeline** ensuring data consistency, referential integrity, and failover reliability across 10+ distributed services — including **automated migration of files to AWS S3** with IAM-governed cross-service access control, covering complex foreign key relationships.
- **Eventlogger Service :** Built a **high-throughput event-driven data warehouse ingesting events** from 10+ services via AMQP/NATS into 40+ structured tables using async DB writes — **decoupling downstream analytics queries from upstream service dependencies**, achieving sub-second ingestion latency and **consolidating multi-service joins into a single source of truth**.

CME Group — World's largest derivatives exchange

SDE Intern

Bangalore, India

May 2023 - Nov 2023

- Re-engineered report and scheduled email generation pipelines for a high-frequency trading data platform, **achieving a 30% throughput improvement by refactoring legacy batch processing logic**.
- Built internal data extraction and transformation tooling that reduced pipeline latency by 40%, enabling faster downstream reporting for trading operations teams.

Projects

KubeSherlock | *Agentic AI, LLM Python, Kubernetes, MCP*

Jul 2026

- **Problem:** Kubernetes pod failures require manual, multi-namespace diagnostic command execution, increasing MTTR.
- Developed an autonomous SRE agent orchestrating a **ReAct reasoning loop over 26 K8s tools via Model Context Protocol (MCP)**, generating root-cause reports with a 100% pass rate across 235 automated tests.
- Implemented a tool-boundary **Security Context layer for namespace ACLs and 100% credential redaction**, coupled with asyncpg schema isolation and a FastAPI Prometheus metrics sidecar.

Mirror-DB | *Kubernetes, PostgreSQL, PgBouncer, Helm Chart*

May 2025

- **Problem:** Production PostgreSQL clusters lose availability during primary failover, causing downtime.
- Designed a HA PostgreSQL cluster on Kubernetes **achieving sub-60s automatic failover, 70–90% connection overhead reduction** via PgBouncer pooling, and read/write splitting for zero-downtime operations.
- Implemented **dynamic service discovery with automated pod labeling**, multi-tier synchronous/asynchronous replication across 4-node cluster, and comprehensive monitoring stack

Technical Skills

Languages: Python, C++, Java, TypeScript, SQL

Databases: PostgreSQL, MongoDB, Oracle (PL/SQL) — schema design, query optimisation, materialized views, data warehousing

Data Engineering: Apache Airflow (DAG authoring), Apache Spark (batch processing), ETL pipeline design

Messaging & Streaming: RabbitMQ, Apache Kafka, NATS (AMQP/JetStream)

Infrastructure & Cloud: Docker, Kubernetes (Helm, pod lifecycle), AWS (S3, EC2, IAM), GitHub Actions

Observability: Prometheus, Grafana

LLM / AI: Anthropic Claude/ OpenAI API, RAG, prompt chaining, structured outputs, self-correction loops

Education

Vellore Institute of Technology, Vellore

Sep, 2020 – July 2024

Bachelors in Computer Science with specialization in Data Science

CGPA - 9.25

Certification

AWS Certified Data Engineer – Associate | Credential ID: d2f78d86511148d89e01d977e84c03bd